Week 9 BA

2024 Sep01 ET

1. [Source Q8; MSQ]
There are 6 business units and you are using the DEA to compare them. You solve the LP for business unit 5. You find from the constraint expression that business unit 5 has obtained an efficiency of 0.7 and business unit 6 has obtained an efficiency of 1 with the optimal weights of business unit 5. Which of the following statements is correct?

(a) Business unit 5 will be efficient

(b) Business unit 6 will be efficient

(c) Business unit 5 will be inefficient

(d) Business unit 6 will be inefficient

[Source ans: b, c]

Comprehension for questions 2-3 (source Q19)

There are 5 business units where we measure the efficiency based on two outputs and one input. You are solving the optimization problem for business unit 4 and you find that the efficiency is 0.8. You find that the dual variables corresponding to the constraints of business units 3 and 5 are non-zero and the dual variables corresponding to the constraints of other units are zero. The dual variables corresponding to the constraints of business units 3 and 5 are 0.5 and 0.3 respectively. You are given the following table where sales and loyal customers are the two outputs. Answer the given subquestions.

	Sales	Loyal Customers
BU 3	12000	100
BU 5	10000	120

2. [Source Q20; SA]
How much is the sales in HCU 4? Hint: Round-off up to 2 decimal places

[Source ans: 11250]

3. [Source Q21; SA]
How many loyal customers must be there in HCU 4? Hint: Round-off up to 2 decimal places

[Source ans: 107 to 108]

2024 Dec22 ET

Comprehension for questions 4-6 (source Q20)

We are interested in understanding the efficiency of 5 Sales Offices. Each of the Sales Offices has an approved budget and a team size for achieving a fixed target of 20,00,000. As a Business Analyst, you must formulate a DEA problem using Linear programming for the Sales Office 3 . Answer subquestions using the Table below.

Sales Office	Budget	Team Size
1	4,00,000	14
2	3,56,000	10
3	6,00,000	8
4	4,90,000	11
5	2,85,000	15

4. [Source Q21; MCQ]

What is the objective function?

(a) $Max: 6{,}00{,}000 * y_{13} + 8 * y_{23}$

(b) $Max: 6{,}00{,}000 * y_{31} + 8 * y_{32}$

(c) $Max: y_{13} * 20{,}00{,}000$

(d) $Max: y_{32} * 20{,}00{,}000$

(e) None of these

[Source ans: c]

5. [Source Q22; MCQ]

What is the type constraint?

(a) $x_{13} * 6{,}00{,}000 + x_{23} * 8 = 1$

(b) $x_{13} * 6{,}00{,}000 + x_{23} * 8 \leq 1$

(c) $x_{31} * 6{,}00{,}000 + x_{32} * 8 \geq 1$

(d) None of these

[Source ans: a]

6. [Source Q23; MCQ]

Which of them is not a constraint for the LP problem pertaining to Sales Office 3?

(a) $y_{13} * 20{,}00{,}000 \leq x_{13} * 4{,}00{,}000 + x_{23} * 14$

(b) $y_{13} * 20{,}00{,}000 \leq x_{13} * 3{,}56{,}000 + x_{23} * 10$

(c) $y_{13} * 20{,}00{,}000 \leq x_{13} * 6{,}00{,}000 + x_{23} * 8$

(d) $y_{13} * 20{,}00{,}000 \leq x_{13} * 4{,}90{,}000 + x_{23} * 11$

(e) $y_{13} * 20{,}00{,}000 \leq x_{13} * 2{,}85{,}000 + x_{23} * 15$

(f) None of these

[Source ans: f]

Comprehension for questions 7-8 (source Q24)

There are 5 business units. There are two outputs and one input under consideration. You are solving the optimization problem for business unit 3 and find that the efficiency to be 0.6. You find that the dual variables corresponding to the constraints of business units 2 and 4 are non-zero and the dual variables corresponding to the constraints of other units are zero. The dual variables corresponding to the constraints of business units 2 and 4 are 0.25 and 0.35, respectively. You are given the following table where sales and number of leads are the two outputs. Based on the above data, answer the given subquestions.

	Output 1	Output 2
BU 2	9000	10
BU 4	6500	12

7. [Source Q25; SA]

How much will the Output 1 in HCU 3? NOTE: For every step performed round off 4 decimal places.

[Source ans: 7541 to 7542]

8. [Source Q26; SA]

How much will the Output 2 in HCU 3? NOTE: For every step performed round off 4 decimal places.

[Source ans: 11.13 to 11.23]

2025 Apr13 ET

9. [Source Q8; MSQ]
There are 7 business units and you are using the DEA to compare them. You solve the LP for business unit 5. You find from the constraint expression that business unit 1 has obtained an efficiency of 1 and business unit 2 has obtained an efficiency of 1 with the optimal weights of business unit 5. Which of the following statements is correct? (choose all that is/are applicable)

(a) Business unit 5 is inefficient

(b) Business unit 1 is efficient

(c) Business unit 5 is efficient

(d) Business unit 2 is efficient

[Source ans: b, d]

10. [Source Q9; MSQ]
In DEA, when can the Linear Programming model be used for calculating the weights of efficiency (weighted outputs/weighted inputs)? (choose all that is/are applicable)

(a) After converting the ratio into the linear objective function

(b) After normalizing the denominator

(c) By setting a constraint on the efficiency of all DMUs to be lesser than or equalto 1

(d) None of these

[Source ans: a, b, c]

Comprehension for questions 11-12 (source Q20)

There are six business units. There are two outputs and one input under consideration. You are solving the optimization problem for business unit 3, and you find that the efficiency is 0.8. You see that the dual variables corresponding to the constraints of business units 2 and 5 are non-zero, and the dual variables corresponding to the constraints of other units are zero. The dual variables corresponding to the constraints of business units 2 and 5 are 0.5 and 0.3, respectively. Based on Table 6, answers the given sub-questions

	Output 1	Output 2
BU 2	9000	10
BU 5	6500	12

Table 6: DEA Table

11. [Source Q21; SA]
How much will the Output 1 in HCU 3?

[Source ans: 8061 to 8064]

12. [Source Q22; SA]
How much will the Output 2 in HCU 3?

[Source ans: 10.5 to 11]

2025 Aug31 ET

Comprehension for questions 13-14 (source Q28)

There are six business units. There are two outputs and one input under consideration. You are solving the optimization problem for business unit 3, and you find that the efficiency is 0.8. You see that the dual variables corresponding to the constraints of business units 2 and 5 are non-zero, and the dual variables corresponding to the constraints of other units are zero. The dual variables corresponding to the constraints of business units 2 and 5 are 0.55 and 0.25, respectively. Based on Table 4, answers the given subquestions. Hint: At every step, round off your answers to 4 decimal places.

	Output 1	Output 2
BU 2	8500	15
BU 5	6500	22

Table 4 DEA Table

13. [Source Q29; SA]
How much will the Output 1 in HCU 3? [Note: Enter your answer in decimal rounded to four decimal places. For example, if your answer is 1000.63545, then enter the answer as "1000.6355"]

[Source ans: 7874 to 7876]

14. [Source Q30; SA]
How much will the Output 2 in HCU 3? [Note: Enter your answer in decimal rounded to four decimal places. For example, if your answer is 1000.63545, then enter the answer as "1000.6355"]

[Source ans: 17.1 to 17.3]

Comprehension for questions 15-17 (source Q31)

We are interested in understanding the efficiency of 5 Sales Offices. Each of the Sales Offices has an approved budget and a team size for achieving a fixed target of 10,00,000. As a Business Analyst, you must formulate a DEA problem using Linear programming for the Sales Office 3 . Answer the given subquestions using the Table below.

Sales Office	Budget	Team Size
1	4,00,000	14
2	3,56,000	10
3	6,00,000	8
4	4,90,000	11
5	2,85,000	15

15. [Source Q32; MCQ]

What is the objective function?

(a) $Max: 3{,}56{,}000 * x_{13} + 10 * x_{23}$

(b) $Max: 3{,}56{,}000 * y_{23} + 10 * y_{23}$

(c) $Max: y_{13} * 10{,}00{,}000$

(d) $Max: x_{13} * 10{,}00{,}000$

(e) None of these

[Source ans: c]

16. [Source Q33; MCQ]

What is the type constraint?

(a) $x_{13} * 6{,}00{,}000 + x_{23} * 8 = 1$

(b) $x_{13} * 6{,}00{,}000 + x_{23} * 8 \leq 1$

(c) $x_{13} * 6{,}00{,}000 + x_{23} * 8 \geq 1$

(d) None of these

[Source ans: a]

17. [Source Q34; MCQ]

Which of them is not a constraint for the LP problem pertaining to Sales Office 3?

(a) $y_{13} * 10{,}00{,}000 \leq x_{13} * 4{,}00{,}000 + x_{23} * 14$

(b) $y_{13} * 10{,}00{,}000 \leq x_{13} * 3{,}56{,}000 + x_{23} * 10$

(c) $y_{13} * 10{,}00{,}000 \leq x_{13} * 6{,}00{,}000 + x_{23} * 8$

(d) $y_{13} * 10{,}00{,}000 \leq x_{13} * 4{,}90{,}000 + x_{23} * 11$

(e) $y_{13} * 10{,}00{,}000 \leq x_{13} * 2{,}85{,}000 + x_{23} * 15$

(f) None of these

[Source ans: f]

2026 May06 ET

Comprehension for questions 18-20 (source Q1)

The following scenario is purely hypothetical A data science company has 5 managers (5 DMUs and k=5). Each manager is allocated the same budget of Rs.3,00,000. The savings achieved (in INR) and the projects completed (successfully) are the outputs.

Manager	Savings	Projects Completed
1	13,50,000	14
2	18,50,000	7
3	32,50,000	14
4	13,24,000	27
5	25,00,000	23

Table 1: Manager Performance

Formulate an optimization problem for Manager 2 to calculate his/her efficiency. Here, the only input is (I_{ik}): the budget to run all the projects. Corresponding to this, there is one input weight (x_{ik}). The outputs are (O_{jk}): Savings, and the Projects Completed. The corresponding output weights are denoted by (y_{jk}).

The LP formulation of the DEA problem for Manager 2.

$$\begin{gathered} Max\ \ a * 18{,}50{,}000 + b * 7 \\ subject\ to\ \ a * 3{,}00{,}000 = 1 \\ a * 13{,}50{,}000 + b * 14 \leq c * 3{,}00{,}000 \\ a * 18{,}50{,}000 + b * 7 \leq c * 3{,}00{,}000 \\ a * 32{,}50{,}000 + b * 14 \leq c * 3{,}00{,}000 \\ a * 13{,}24{,}000 + b * 27 \leq c * 3{,}00{,}000 \\ a * 25{,}00{,}000 + b * 23 \leq c * 3{,}00{,}000 \\ Decision\ variables:\ a, b, c \geq 0 \end{gathered}$$

Based on the data provided, answer the given sub-questions.

18. [Source Q2; MCQ]
Fill in the correct value a.

(a) x12

(b) x22

(c) y21

(d) y12

[Source ans: d]

19. [Source Q3; MCQ]
Fill in the correct value b.

(a) y12

(b) x22

(c) x21

(d) y22

[Source ans: d]

20. [Source Q4; MCQ]
Fill in the correct value c.

(a) x22

(b) x12

(c) y21

(d) y22

[Source ans: b]

Comprehension for questions 21-22 (source Q8)

The following scenario is purely hypothetical Upon solving the optimization problem for business unit 2, you find the efficiency is 0.8. You also find that the dual variables corresponding to the constraints of business units 5 and 3 are non-zero, and the dual variables corresponding to the constraints of other business units are zero. The dual variables corresponding to the constraints of business units 5 and 3 are 0.3 and 0.5, respectively.

	Savings	Projects Completed
BU 5	25,00,000	23
BU 3	32,50,000	14

Based on the given data, answer the given sub-questions.

21. [Source Q9; SA]
How much will the Savings in HCU 2? (Note: Enter your answer in decimal values rounded to two decimal places. For example, if your answer is "1.235", then enter the answer as "1.24")

[Source ans: 2968749 to 2968751]

22. [Source Q10; SA]
How much will the Projects Completed in HCU 2? (Note: Enter your answer in decimal values rounded to two decimal places. For example, if your answer is "1.235", then enter the answer as "1.24")

[Source ans: 17.1 to 17.8]

23. [Source Q11; MSQ]
In DEA, when can the Linear Programming model be used for calculating the weights of efficiency (weighted outputs/weighted inputs)?

(a) After converting the ratio into the linear objective function

(b) We solve a non-linear optimization problem

(c) After normalizing the denominator

(d) By setting a constraint on the efficiency of all DMUs to be lesser than or equal to 1

(e) All of these

[Source ans: a, c, d]